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A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

R26-SE-026 | Progress Presentation

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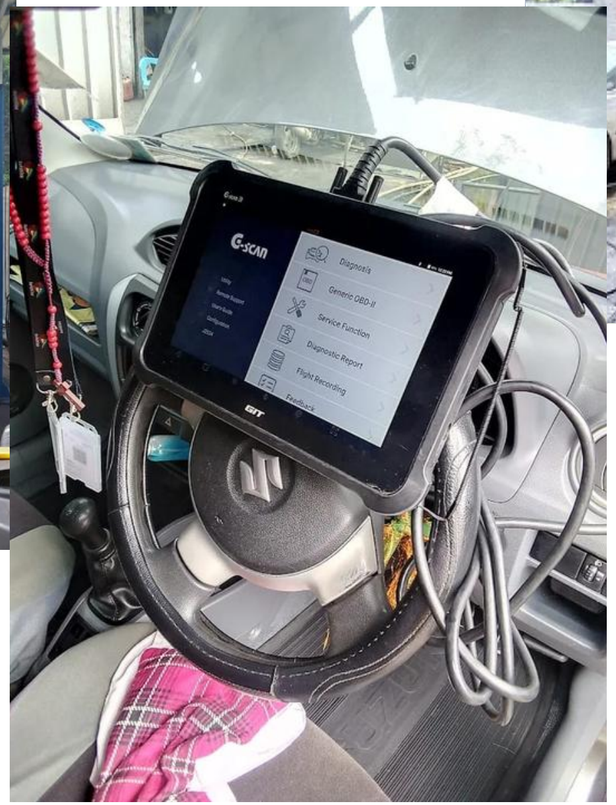
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Proposal Feedback & Our Response

Action Item	What we did
Proper dataset validation	Collected Data from mechanics and found specific datasets from Kaggle Cross-checked NTC / MoH / data.gov.lk Reproducible: generator independent of scorer
Domain expert verification of parameters	Questionnaire, service types and scenarios validated against mechanics and discussions going with Automobile Association of Ceylon. Anchored in HCM 7th Ed · LWR shockwave · Koliou 2024 (5 factors)
Clear target users + practical outputs	Stranded drivers (B2C), Service providers, fleet operators (B2B), Insurance + Government (enterprise) Mapped Users with FR. Output classified per consumer

Field Visit and Client Meetup









A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka



Problem and Impact:

- Vehicle breakdowns rely on manual calls, causing long waits, provider mismatches, fraud-prone claims, traffic congestion, reactive maintenance, and no data privacy compliance.

Who will benefit from the Solution:

- Private vehicle owners facing delayed emergency response
- Licensed insurance companies losing revenue to fraudulent claims
- Registered repair centers and garage networks
- Roadside assistance providers and towing services
- Daily commuters affected by incident-induced traffic delays

Research Objective & Sub-Objectives:

- Unified smart platform integrating four intelligent modules for end-to-end roadside assistance and automotive service management in Sri Lanka.
  - Uncertainty-Aware Dispatch Optimization
  - Geo-Intelligence & Traffic Impact Analysis
  - Predictive Maintenance for Service Management
  - Intelligent 3D Accident Reconstruction

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Component: Dispatch Optimization & Smart Recovery

Contribution

Core decision making engine — determines what service an incident actually needs and selects the optimal service provider/ assistance minimizing mismatch rates and traffic congestion.

Novelty

First framework to treat roadside dispatch with **service type uncertainty**, considering external traffic and cost for dispatch optimization and adaptive learning.

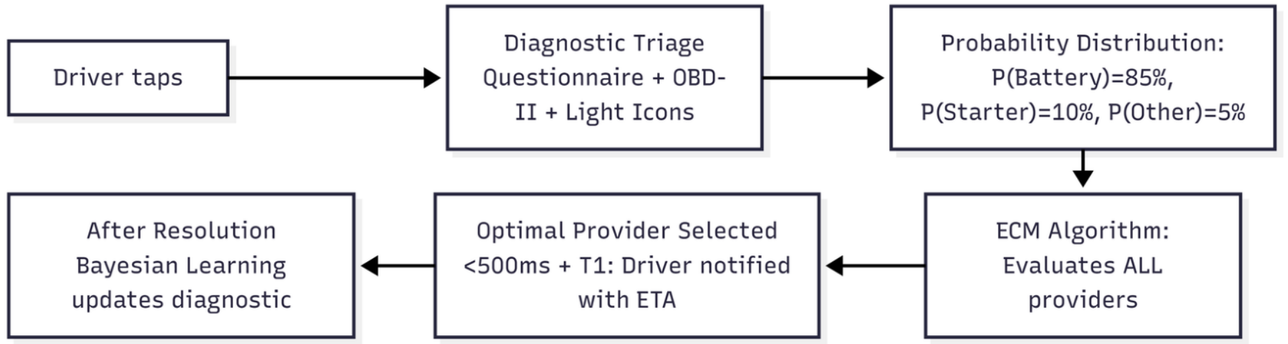
Dispatch Optimization & Smart Recovery

Problem

Drivers rely on Manual phone calls for roadside dispatch and find difficult to connect to the right service provider for their need and may end up in wrong service provider, which costs time and energy.

Solution Addressed

The prototype addresses the core problem of manual, error-prone roadside dispatch by introducing an Uncertainty-Aware Dispatch Optimization



User Requirements Addressed

- System compatible for both manual vehicles and sensor vehicles.
- Find the uncertain incident and let the user know the issue.
- Address known issues quickly and add more common incidents.
- Dispatch to the most appropriate service provider.
- Reduce the mismatch rate and dispatch to the most accurate service provider.
- Receive real-time provider assignment with ETA and diagnostic explanation.

Dispatch Optimization & Smart Recovery

Design Excellence/ Contribution

- First roadside dispatch system modeling service-type uncertainty as a probability distribution before provider selection
- Multi-modal diagnostic triage combining questionnaire, OBD-II telemetry, visual icons, and audio samples
- Original ECM algorithm: Σ P(type)×Cost + λ×Traffic×Time ÷ Trust — accounts for mismatch risk and traffic externality
- PDPA-compliant Decision Tree selected over Random Forest (84% vs 86%) — single traceable path, fully auditable, exportable as JSON
- Bayesian learning loop with decaying learning rate refining leaf-level probabilities from real incident outcomes
- Three-tier graceful degradation: OBD-enhanced → questionnaire-only → Bayesian-refined

User Feedback on Prototype

- Dashboard light icons preferred over text descriptions
- Mechanics confirmed the diagnostic question flow matches their real-world assessment sequence
- Providers want clear job cards showing pre-diagnosis and required equipment before accepting
- Users suggested adding Sinhala/Tamil language support for wider accessibility in the future.
- Identify dashboard warning lights through visual icon selection



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Component: Geo-Intelligence & Traffic Impact Analysis

Contribution: Spatial awareness layer — provides traffic intelligence to dispatch and congestion visualization to traffic authorities

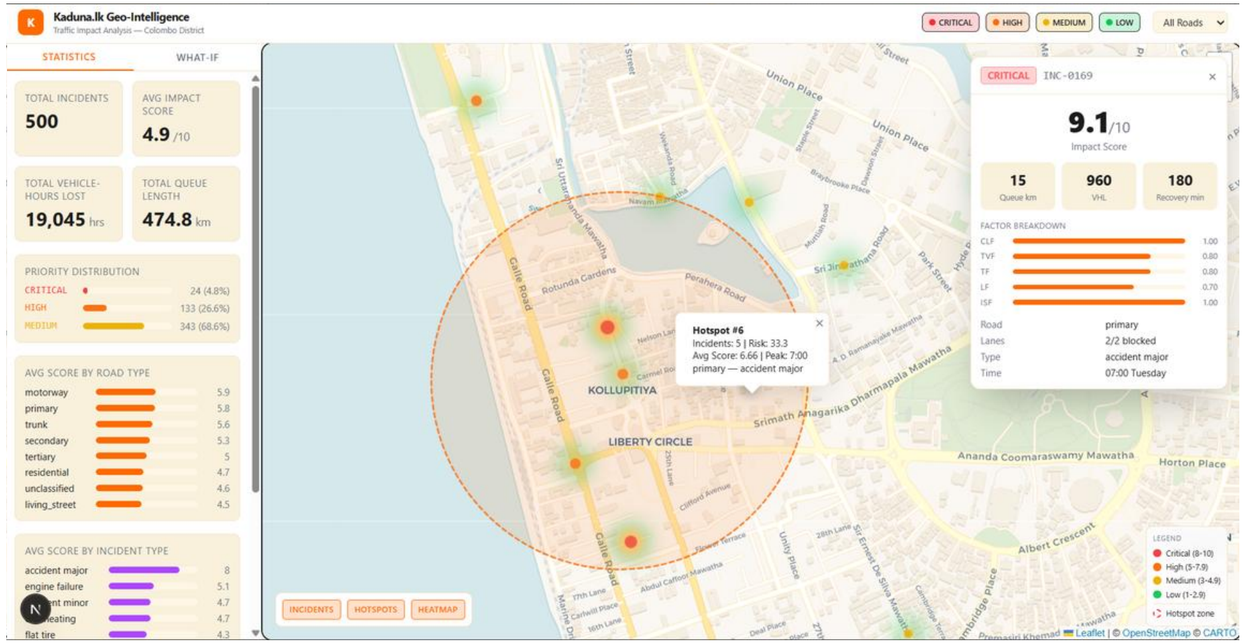
Novelty: First incident impact scoring model calibrated for Sri Lankan road conditions, validated against SUMO microscopic traffic simulation

Geo-Intelligence and Traffic Impact Analysis— Methodology

- **Sub-Problem**
  - All incidents treated equally — no traffic-impact awareness
  - No prediction of how a breakdown spreads through nearby traffic
- **Solution**
  - 5-factor Impact Score → priority tier (CRITICAL / HIGH / MED / LOW)
  - Shockwave-based queue & VHL prediction
  - Hotspot map (KDE + DBSCAN)
  - REST score-API (FastAPI + OpenAPI 3.1) → contract published for dispatch integration
- **User Requirements (mapped to roles)**
  - Operator → priority signal · factor breakdown · hotspot overlay (FR-01, 02, 06, 10)
  - Dispatcher → queue · VHL · recovery prediction · REST score-API (FR-03, 08)
  - Fleet mgr → what-if simulator for corridor risk (FR-07)
- **Design Excellence**
  - Weights learned from SUMO simulation, not hand-picked
  - Interpretable formula → defensible dispatch decisions

Geo-Intelligence and Traffic Impact Analysis — Implementation

- **Prototype**
  - Scored Colombo dataset
    - SUMO pipeline
    - scoring API
    - Next.js dashboard
  - Map
    - severity heatmap
    - hotspot overlay
    - what-if simulator



- **Feedbacks:**
  - Supervisor: methodology flow accepted; IEEE citations confirmed
  - Dispatch component (Janukshan): REST contract endorsed for PP2 integration
  - Internal walkthrough: what-if simulator UX + factor breakdown praised
  - Scheduled post: traffic-authority practitioner walkthrough

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Predictive Vehicle Maintenance

Problem & Solution



Vehicle Data **(OBD-II)** & behaviour

↓

RUL

↓

The Right Repair

The Solution Addresses the Sub-Problem/ Prototype

OBD-II data is collected from the ELM327 adapter and sent to backend with mobile sensor data.

The backend stores the trip data and calculates the average vehicle data.

The data is sent to ML models to predict the RUL in kilometers.

Another Random Forest ML model is used to predict the maximum usage (km) of vehicle parts and calculate the RUL percentage.

Smart recommendations are generated from the marketplace based on the predicted RUL.

Live testing was conducted using a real vehicle connected to an ELM327 OBD-II adapter.

Addressed Requirements

- Manage Vehicles and their service data ✓
- Get Vehicle Live data and analyse them ✓
- Predict RUL for each vehicle part
  Engine, Break, Tyre, Battery ✓
- choose the best and acurate models for predictiom ✓
- Give RUL as a Percentage ✓
- Smartly suggest the best service provider ✓

Design Excellence / Contribution

- **Multi-modal IoT sensor fusion** — OBD-II telemetry combined with IMU motion data
- **Dual-layer ML architecture** — 5 models operating as a single pipeline
- Per-component prediction windows anchored to service records
- Context-adaptive health scoring calibrated to Sri Lankan operating conditions

User Feedback on Prototype

- Give recommendation to user to adjust their driving behaviour.
- Service records are need to be verified to avoid fraud.
- Give Remaining Useful life in Time instead of Kilometers

Design Excellence / Contribution

- **Multi-modal IoT sensor fusion** — OBD-II telemetry combined with IMU motion data
- **Dual-layer ML architecture** — 5 models operating as a single pipeline
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User Feedback on Prototype

Give recommendation to user to adjust their driving behaviour.

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Component

Intelligent 3D Accident Claim System

Contribution: A mobile-first 3D accident scene reconstruction system using guided photo capture, low-light enhancement, and volumetric fraud detection to replace unreliable 2D photo submissions.

Problem

Non-standard accident imaging, agent delays, claim fraud, poor nighttime visibility, rural access limitations, and inaccurate 2D-only damage assessment.

Intelligent 3D Accident Claim System

The Solution Addresses the Sub-Problem/ Prototype

The app will guide the user to take pictures around the car in a standard way.

The app will guide the user to submit vehicle driver data as required by the insurance companies in Sri Lanka

After completing the checklist, the user can submit the photos

Enhance the low-light images and show them to the Insurer with other photos of the accident

To generate the 3D mesh model using the uploaded 2D images of the accident.

Intelligent 3D Accident Claim System

Addressed Requirements

No agents in rural areas

Stop claim recycling

Only upload application photos

Submit the photos quickly with guidance

Validate the vehicle driver data

Enhance the low-light images

Generate the 3D model

Intelligent 3D Accident Claim System

Design Excellence / Contribution

- **Keep the data persistent** — Even if the user accidentally closes the app, the data will not delete
- **Sequential uploads** — The photos will be uploaded in different steps
- **Guide the user to complete the steps** — capture the images, and to verify the user in one clear checklist

User Feedback on Prototype

- The UI looks clear to the user and can be read without spectacles.
- Give the user permission to video a small clip in case of a lower-body collision
- Ask what information that is needed to verify the Driver of the car
- Make the user to upload verification photos at the same time

Commercialization Plan

- **Target Market:**
  - Private vehicle owners, Fleet operators, insurance companies, repair centres, Traffic authorities
- **Value Proposition:**
  - First unified smart platform for roadside assistance in Sri Lanka — intelligent dispatch, traffic intelligence, predictive maintenance, 3D fraud prevention, PDPA compliance
- **Budget:**
  - Software Costs and Marketing - LKR 80,000–120,000
- **Revenue Model:**
  - Freemium for individual drivers, B2B subscription for fleet/insurance, Government licensing, Marketplace commission, Advertising
- **Competitive Advantage:**
  - First ever in Sri Lankan market
  - Validated through simulation & Integrated platform approach
- **Intellectual Property Rights**:
  - Proprietary algorithms eligible for protection - Academic publications planned
  - Software Copyrights and Patent
  - Trademark for Branding

Commercialization Plan

Customer Persona

- Private vehicle owners, Fleet operators, insurance companies, repair centres, Traffic authorities

Market

- Sri Lankan vehicle users
- Freemium for individual drivers, B2B subscription for fleet/insurance, Government licensing, Marketplace commission, Advertising

What Is Unique in your solution

- Integrated ecosystem connecting drivers, garages, insurers, and authorities

How the cost be covered

- Monthly/annual B2B subscriptions for fleet operators and insurance companies

Thank You

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Questions & Discussion